

Model Practical

Course code: CSA0704

Course name: Computer Networks

Reg. no: 1924260025

①. AIM:

To build a network in Packet Tracer and configure an extended Access control list (ACL) to permit only HTTP traffic from a specific subnet to a web server, while blocking all other traffic.

Procedure:

• Packet Tracer:

• No of systems: 1 PC (client), 1 server (webserver),

1 router, 2 switches

• Hardware: Cisco Router, switches & copper straight-through cables

• IP/SM:

• PC : 192.168.1.10 / 255.255.255.0 (gateway: 192.168.1.1)

• Server : 192.168.2.10 / 255.255.255.0 (gateway: 192.168.2.1)

Configuration steps:

1. Assign IPs to the PC, server and router interfaces

2. Create Extended ACL on the router

```
access-list 100 permit tcp 1
```

```
access-list 100 deny ip any.
```

3. Apply to interface.

```
ip access-group 100 in
```

Sample O/p & screenshot:

- Command Prompt Output:

ping 192.168.2.10 shows

"Destination host unreachable"

- Web browser Output:

Navigating to [http://192.168.2.10] (http://192.168.2.10)

Successfully loads the web page.

Result:

The Extended ACL was successfully configured & verified in packet tracer to allow only HTTP traffic while blocking all other traffic.

AIM:

To design and justify a WAN topology connecting 6 remote offices to a central head office in Packet Tracer.

Procedure:

Packet Tracer:

- No. of System: 7 Routers (1 central, 6 remote)
- Hardware: Routers with serial interfaces, serial DCE cables.
- Topology choice: Hub and spoke (star topology)
- Justification: Cost effective (only requires 6 links instead of a full mesh) and allow centralized management at the head office.
- Steps: Place the central router, surround it with 6 remote routers and connect them all to the center using serial cables.

Result: A hub and spoke WAN topology was successfully designed and justified.

30

AIM: To capture and analyze Network Time Protocol (NTP) packets using Wireshark.

Procedure:

- Wireshark:

- Protocol: NTP (UDP Port 123)

- Steps:

1. Start Wireshark capture on the active network interface.

2. Force the PC to sync its data / time settings.

3. Type ntp in the display filter

4. Inspect the UDP header and NTP payload fields (Flags, Stratum, Timestamps).

Result: NTP packets were successfully captured, filtered and analyzed using Wireshark.

4.

AIM:

To implement and demonstrate the cyclic redundancy check (CRC) for a given data frame using a C program.

Procedure:

- C Program:

- Filename: crc.c

- Functions used: main(),

- printf(), scanf(),

- strlen(), strcpy(),

- strcpy().

- Compile & Execute:

- gcc crc.c -o crc then ./crc

Sample O/P & screenshot:

Enter data: 10101010

Enter gen: 1011

checksum: 011

Transmitted: 10101010011

Result: The CRC error detection algorithm was successfully implemented and verified using a C program.